



Ulimpiad n Tefriqt n Ugafa n Tselkimt Tamenzut 2025

Tiles

Tilist n wakud: 2 secondes

Tilist n tkatut: 256 MB

Tufid yiwet n tfelwit taqdimt yescan $n \times m$ n yimekkuzen, anda $1 \leq n \leq 2$. Kra n yimekkuzen ttwaqeffen, wiyad ferɣen akk. Nezmer ad-nessers azru n Ddumin (asrem lqis-is 1×2 neɣ 2×1) ɣef sin n yimekkuzen iferɣen yerna neɣtɣen. Mi ara nesses azru, Imekkuzen i y emmer ad uɣalen ttwaqeffen. Neqqar-d belli agraw n yizra **yelha** i yiwet n tfelwit, ma yella ur nezmir ara ad nernu izra-nniɣen mi ara nesses akk izra.

Ma nefka-ak-d tafelwit, aɣal n igrawen yelhan n yizraq i yellzn ? Eɣseb modulo $10^9 + 7$. Sin n yizra mgaraden ma yella umekkuɣ i yeemmer yiwen yerna wayed u t-yeemmer ara. Sin n igrawen mgaraden ma yella uzru deg yiwen n ugraw ulac-it deg wayed.

Anekcum

Ajerriɣ amnzu deg-s sin n uɣtunen n akked m ($1 \leq n \leq 2$, $1 \leq m \leq 10^5$).
Ijerriɣen n i yellan membaed sean m n isekkilen yal yiwen, yal asekkil ad yili d '' neɣ '#'.
Asekkil j deg ujerriɣ i ad yili d '' ma yella amekkuɣ (i, j) yefreɣ,
neɣ '#' ma yella amekkuɣ (i, j) yettwaqfel.

Tuffɣa

Suffeɣ-d yiwen n uɣtun - uɣtun n yigrawen yelhan modulo $10^9 + 7$

Tiɣuriwin

- $1 \leq n \leq 2$
- $1 \leq m \leq 10^5$

Iluɣma imeɣyanen

Alaɣmu	Tineqqiɣin	Tiɣuriwin
1	6	$m \leq 2$
2	14	$n = 1$
3	12	$m \leq 7$
4	8	$3 \mid m$, i akk $j = 3k$ (k seg 1 ɣer $\frac{m}{3}$) imekkuzen $(1, j - 2)$, $(1, j - 1)$ akked $(2, j - 2)$ ferɣent, tiyyad akk ala.
5	12	Akk imekkuzen ferɣen
6	18	$n \leq 10^3$
7	30	-

Imedyaten

Anekcum 1

2 2

..

..

Tuffγa 1

2

Anekcum 2

2 3

##.

##.

Tuffγa 2

1

Anekcum 3

1 8

...#...

Tuffγa 3

4

Asegzi

- Deg umedya amenzu llan sin n wagraŵen yelhan : yiwen yes3a sin n yizra bedden, wayed yesεa sin n yizra telqen.
- Deg umedya wis sin, agraw yelhan d agraw ur yesεin ula.